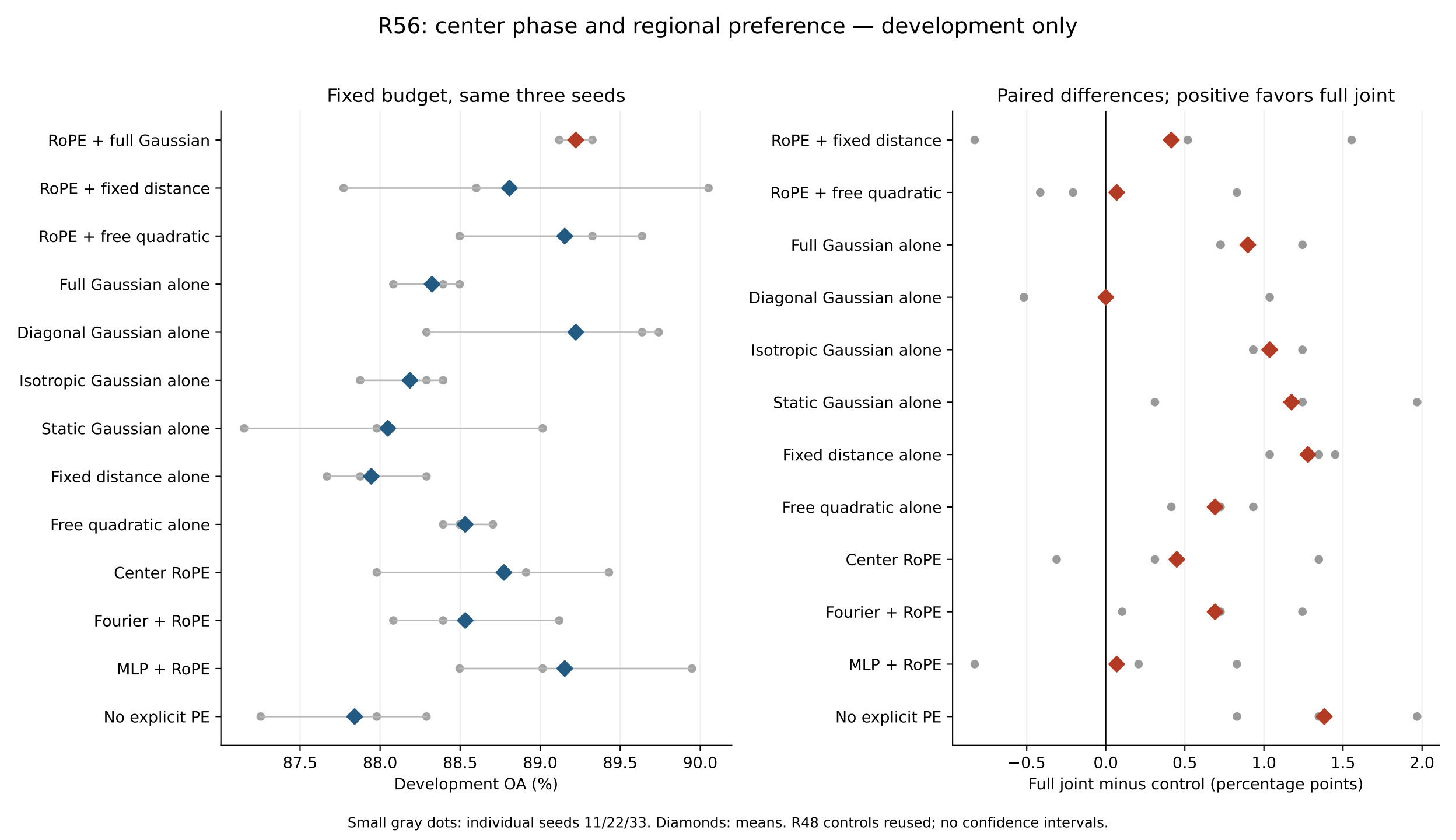
R56: center phase and regional preference — development only
Fixed budget, same three seeds
RoPE + full Gaussian
RoPE + fixed distance
RoPE + free quadratic
Full Gaussian alone
Diagonal Gaussian alone
Isotropic Gaussian alone
Static Gaussian alone
Fixed distance alone
Free quadratic alone
Center RoPE
Fourier + RoPE
MLP + RoPE
No explicit PE
87.5
88.0
88.5
89.0
89.5
90.0
Development OA (%)
Paired differences; positive favors full joint
RoPE + fixed distance
RoPE + free quadratic
Full Gaussian alone
Diagonal Gaussian alone
Isotropic Gaussian alone
Static Gaussian alone
Fixed distance alone
Free quadratic alone
Center RoPE
Fourier + RoPE
MLP + RoPE
No explicit PE
−0.5
0.0
0.5
1.0
1.5
2.0
Full joint minus control (percentage points)
Small gray dots: individual seeds 11/22/33. Diamonds: means. R48 controls reused; no confidence intervals.